ELEC70048
EE9-CS5-1
EE9-SC3
EE9-AML2-01

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE PART IV: MEng and ACGI

PROBABILITY AND STOCHASTIC PROCESSES

Tuesday, 30 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions. All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Information for students

Each of the four questions has 25 marks.

The Questions

1. Random variables.

a) There are 17 fenceposts around the perimeter of a field, exactly 8 of which are rotten. Show that irrespective of which these 8 are, there necessarily exists a run of 7 consecutive posts at least 4 of which are rotten.

[10]

b) Let random variable X take non-negative integer values.

i) Show that

$$E[X] = \sum_{n=0}^{\infty} P(X > n) \qquad (1.1)$$

[8]

ii) An urn contains 20 red and 10 blue balls. Draw the balls uniformly at random without replacement until the first blue ball is seen. Using the formula (1.1) given above, find the expected number of draws (until the first red ball is seen). [7]

2. Estimation.

a) A random variable X has an exponential distribution with parameter $\lambda = 2$

$$f(x) = \lambda e^{-\lambda x}.$$

We wish to estimate λ from $n = 20$ i.i.d. samples of X. Compute the Cramer-Rao lower bound $\mathrm{Var}[\hat{\lambda}] = \frac{1}{I(\lambda)}$ where the Fisher information

$$I(\lambda) = -E\left[\frac{\partial^2}{\partial \lambda^2} \ln f_X(x_1, \cdots, x_n; \lambda)\right]$$

where $f_X(x_1, \cdots, x_n; \lambda)$ is the joint pdf.

[10]

b) Consider a fading channel in wireless communications, where the channel gain $Y(n)$ (n is the time index) has autocorrelation function

$$R_Y(n) = \begin{cases} 1 & n = 0 \\ 0.1 & n = 1 \\ -0.2 & n = 2 \end{cases}$$

Suppose we wish to predict $Y(n+1)$ from $Y(n)$, $Y(n-1)$, …, $Y(1)$ using the linear MMSE estimator

$$Y(n+1) = \sum_{i=1}^{n} c_i Y(i)$$

i) Compute the coefficient and mean-square error of the first-order linear MMSE estimator, i.e., $n = 1$. [6]

ii) Compute the coefficients and mean-square error of the second-order linear MMSE estimator, i.e., $n = 2$. [9]

3. Random processes.

a) Consider the random process $X(n) = \cos(2\pi nU)$ for $n = 1, 2, 3, \ldots$ where U is uniformly distributed on interval [0, 1]. Find the marginal probability density function of random variable $X(n)$ for a given n. [10]

b) Three girls A, B and C are playing table tennis. In each game, two of the girls play against each other and the third girl does not play. The winner of any given game n plays again in game $n + 1$, against the girl who did not play game n. Assume the probability that girl x will beat girl y in any game that they play against each other is always 1/2.

i) Represent this process as a discrete-time Markov chain by defining the possible states and constructing the transition matrix. [5]

ii) Determine the probability that two girls B and C, who play each other in the first game, will play each other again in the 5th game. [10]

4. Markov chains and martingales.

a) Historically, martingales refer to the "doubling the stake" strategy in gambling, where the gambler doubles the bet on every loss, till the almost sure win occurs eventually at which point the entire loss is recovered. Let the gambler start with a bet £1 at time $n = 0$.

i) If the game is fair, show the gambler's fortune $\{X_n\}$ forms a martingale. [3]

ii) Suppose the gambler wins for the first time at the Nth play. Find the probability distribution of N. [3]

iii) By this time, what is the expectation of the gambler's loss? [4]

b) Consider a symmetric random walk on the two-dimensional grid $\{(x, y): x, y \in 0, \pm 1, \pm 2, \dots\}$. This Markov chain is a sequence $\{X_n\}$ where $P\{X_{n+1} = X_n + \epsilon\} = \frac{1}{4}$ and the vector $\epsilon \in \{(\pm 1, 0), (0, \pm 1)\}$. Suppose the chain starts at the origin $X_0 = (0,0)$.

i) Derive the probability $P\{X_{2n} = (0,0)\}$. [5]

ii) Show that the origin is a recurrent state. [10]

Hint: Being recurrent requires $\sum_n P\{X_{2n} = (0,0)\} = \infty$. Stirling's formula $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$.